Spring 2017 PHY102: Introduction to Physics-II 29/04/2017

End-semester Examination (Full Marks = 35)

Department of Physics Duration: 2hrs30min Shiv Nadar University

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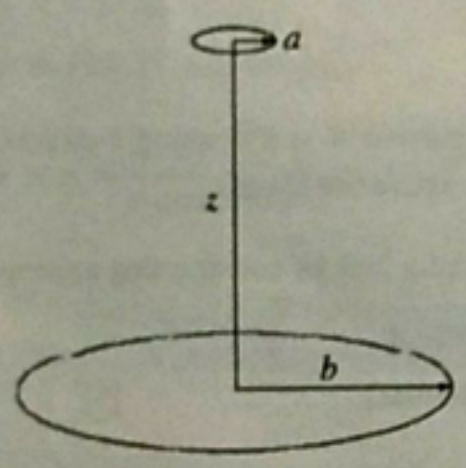
1. Choose the correct option giving **proper reasons** supported by **appropriate calculations** for your choice. *NO marks will be awarded without proper argument*. In case of multiple attempts to the same question, only the first attempt would be regarded as valid. [11 marks]

(a) A small loop of wire of radius a is kept close to a large current carrying loop of wire of radius b ($b >> a$) (figure below). The magnetic flux passing through the small loop due to current I in the larger loop is $\emptyset$. Now, at the same time, if the same magnitude of current flows in the smaller loop but in a reverse direction (that is $-I$), what is the flux passing through the larger loop due to the current in the smaller loop?



(i) $(\frac{a^2}{b^2})\emptyset$ (ii) $(\frac{b^2}{a^2})\emptyset$ (iii)) $-\emptyset$ (iv) $\emptyset/z$ [2]

(b) In the above question, if loops are allowed to move due to their mutual electromagnetic forces, what is the effect on current flowing in the larger loop?

(i) increase (ii) decrease (iii) no change [2]

(c) A single-slit diffraction pattern is formed on a distant screen. Assuming the angles of diffraction involved are small, by what factor will the width of the central bright spot on the screen change if (a) the wavelength is doubled, (b) the slit width is doubled. [2]

(d) Find the average energy density ($u = \frac{\epsilon_0 E^2}{2} + \frac{B^2}{2\mu_0}$) and flux density ($\vec{S} = \frac{\vec{E}\times\vec{B}}{\mu_0}$) of an electromagnetic wave moving in free space and having vector potential $\vec{A} = \cos(kz - \omega t)\hat{i}$. [3]

(e) Two coherent microwave sources both produce waves of wavelength 1.50 cm. The sources are located in the $z = 0$ plane, one at $x = 0$, $y = 15.0$ cm and the other at $x = 3.00$ cm, $y = 14.0$ cm. If the sources are in phase, find the

difference in phase between these two waves for a receiver located at the origin. [2]

2.

(a) A square loop of sides 'a' centered at origin and in $y-z$ plane carries a steady current 'I' in anticlockwise direction. The loop is kept in the region of a magnetic field $\boldsymbol{B} = kz\hat{x}$. What is the magnitude and direction of the net force acting on the loop? **[3]**

(b) Express Ampere's law in differential form. How do you convert the differential form into an integral form? **[1.5]**

A long solenoid having 'n' turns per unit length carries a steady current 'I'. Using integral form of Ampere's law ***only***, find the magnetic field inside the solenoid, assuming magnetic field approaches zero as one goes outside the solenoid. **[1.5]**

3.

(a) An infinite sheet of conductor in $x-y$ plane carries a uniform surface current $\boldsymbol{K} = K\hat{x}$ along x-direction. A current carrying **circular** loop of radius R carrying a steady current I is placed above the sheet.

Find the vector potential of the current carrying loop in the region $s < R$ and $s > R$ by constructing appropriate Amperian loops.

[**Hint:** You can use the integral relation between magnetic flux and vector potential.] **[3]**

(b) From the theory of multipole expansion of vector potential due to one dimensional current distribution in arbitrary closed loop, express the vector potential $\boldsymbol{A}$ at a distance r (distance with respect to origin of the current loop) in terms of magnetic dipole moment $\boldsymbol{m}$. **[1]**

How do you calculate the *'approximate'* magnetic field $\boldsymbol{B}$ (at distances far away from the origin) from the above vector potential considering that a magnetic dipole moment is placed at **origin** and pointing along **z-axis**? Express your answer in terms of spherical polar co-ordinates. **[2]**

4.

(a) Consider a circular loop of wire with radius R in the $x-y$ plane centered at origin and carrying a steady current I running counter-clockwise (as viewed from above z-axis). Using ***strictly*** your answer for the magnetic field $\boldsymbol{B}$ in **Q3(b)** above, find the magnetic field on the points of z-axis above and below the plane of current carrying wire. **[3]**

(b) A spherical shell of radius 'R' and carrying surface charge density σ rotates with angular velocity ω about the vertical axis in anticlockwise manner. Find the dipole moment of the spinning spherical shell.

Show that the vector potential at points $r > R$ is given by $\boldsymbol{A}_{dip} = \frac{\mu_0 \sigma \omega R^4}{r^2} sin\theta(\hat{\varphi})$. **[3]**

5.

(a) Using the Maxwell equations in free space, derive the equation $\nabla^2 E = \mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2}$ and $\nabla^2 B = \mu_0 \epsilon_0 \frac{\partial^2 B}{\partial t^2}$. From this equation argue that the electromagnetic wave move with the speed of light. **[3]**

(b) Consider the case of a double slit-diffraction with the slit width $b = 8.8 \times 10^{-3}\ cm$, distance between two slits $d = 7 \times 10^{-2}\ cm$ and $\lambda = 6.328 \times 10^{-5}\ cm$. How many interference minima will occur between two diffraction minima on either side of central maximum? Calculate the fringe width if the screen in placed at a distance 15 feet on the focal plane of lens placed between the diffraction aperture and screen. **[3]**

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Important formulae

Spherical Polar Co-ordinates:

$$\nabla \times \boldsymbol{v} = \frac{1}{r\sin\theta}\left[\frac{\partial}{\partial\theta}(\sin\theta\, v_\varphi) - \frac{\partial v_\theta}{\partial\varphi}\right]\hat{\boldsymbol{r}} + \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial v_r}{\partial\varphi} - \frac{\partial}{\partial r}(r v_\varphi)\right]\hat{\boldsymbol{\theta}} + \frac{1}{r}\left[\frac{\partial}{\partial r}(r v_\theta) - \frac{\partial v_r}{\partial\theta}\right]\hat{\boldsymbol{\varphi}}$$

$$\nabla \cdot \boldsymbol{v} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\, v_\theta) + \frac{1}{r\sin\theta}\frac{\partial v_\varphi}{\partial\varphi}$$

$$\hat{\boldsymbol{z}} = cos\theta\hat{\boldsymbol{r}} - sin\theta\hat{\boldsymbol{\theta}}$$

Cylindrical co-ordinates:

$$\nabla \times \boldsymbol{v} = \left[\frac{1}{s}\frac{\partial v_z}{\partial\varphi} - \frac{\partial v_\varphi}{\partial z}\right]\hat{\boldsymbol{s}} + \left[\frac{\partial v_s}{\partial z} - \frac{\partial v_z}{\partial s}\right]\hat{\boldsymbol{\varphi}} + \frac{1}{s}\left[\frac{\partial}{\partial s}(s v_\varphi) - \frac{\partial v_s}{\partial\varphi}\right]\hat{\boldsymbol{\varphi}}$$

$$\nabla \cdot \boldsymbol{v} = \frac{1}{s}\frac{\partial}{\partial s}(s v_s) + \frac{1}{s}\frac{\partial v_\varphi}{\partial\varphi} + \frac{\partial v_z}{\partial z}$$

$$\nabla \times \nabla \times \boldsymbol{v} = \nabla(\nabla.\boldsymbol{v}) - \nabla^2\boldsymbol{v}$$

Fundamental Constants

$\mu_0 = 4\pi \times 10^{-7} N/A^2,\ \epsilon_0 = 8.85 \times 10^{-12} C^2/Nm^2,\ c = 3.0 \times 10^8 m/s$